SJ360 Systems Handbook

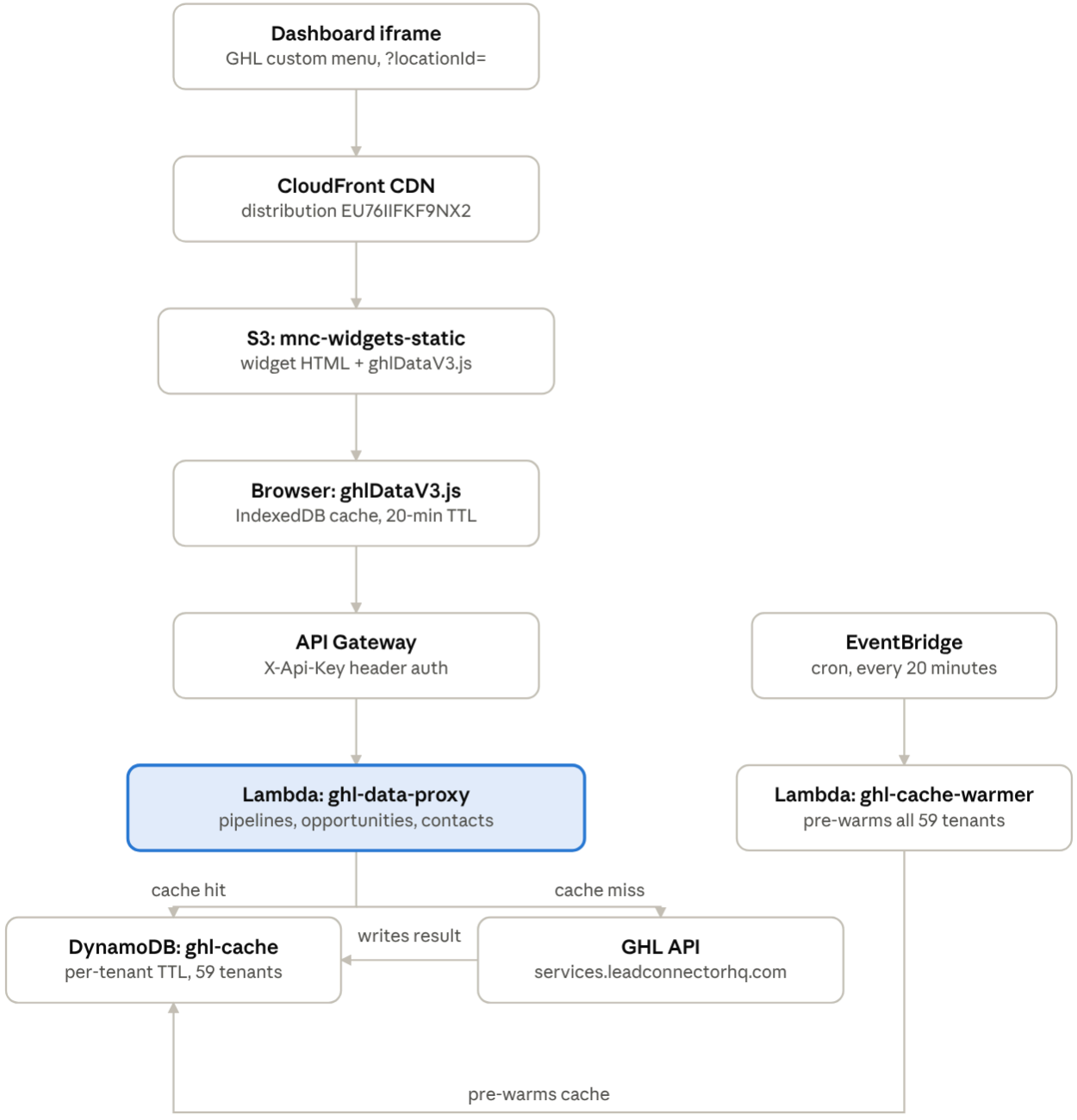
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A simple guide to this whole project: every chart and report screen we built, where the computer files for them live, and how it all connects behind the scenes. Written in plain language — you don't need any coding background to follow it.

How It All Connects

A dashboard load flows iframe to CDN to Lambda to GHL

Cached at the browser, at the Lambda proxy, and in DynamoDB



How a report page gets its numbers, step by step

In plain words, here's what happens when someone opens a report:

1. Someone opens their GHL dashboard and clicks a tab. That tab is really a small window showing one of our report pages.
2. That report page is a file sitting in Amazon's online storage — think of it as a website, except hosted by Amazon instead of us.

3. The page then asks a second Amazon service for the real numbers (like "how many leads came in this week").
4. That service checks if it already saved a copy of the answer recently (in the last 20 minutes). If yes, it hands back that saved copy right away — much faster than asking GoHighLevel (GHL) again.
5. If there's no recent saved copy, it asks GHL directly, saves a fresh copy for next time, and sends the answer back.
6. A separate automatic helper quietly refreshes the saved copies for all 59 companies every 20 minutes, so most of the time nobody has to wait at all.

Every Chart And Report, Explained Simply

All of these are small web pages that show up as tabs inside a company's GHL dashboard. They all live in the same online storage locker (on Amazon), so updating one means replacing the file there. Below, each one's job is explained in plain words, grouped by the folder it's saved in.

Bar Charts (saved in the `bar` folder)

- **Leads by Hour** (`bar_indexedDB.html`) — Shows how many leads come in during each hour of the day, and whether that's more on weekdays or weekends. You can filter by sales pipeline, date range, and lead source.
- **Leads by Hour — OIB North version** (`bar_indexedDB_oib.html`) — The same chart, set up specifically for OIB North: one less filter, and the "Source" list only fills in once you've picked a "Source Category" first.
- **Leads by Day** (`bar_indexedDB_days.html`) — Same idea as "Leads by Hour," but counts leads per calendar day instead of per hour.
- **Leads Enquiry Stats** (`leads_enquiry_stats.html`) — Lets you compare two time periods side-by-side (like this month vs last month) to see if things are improving.
- **Source Analytics** (`source_last_entry.html`) — Ranks where leads come from (Facebook, website, etc.) by how many leads each source brought in, and shows when each source last brought one in.
- **Deal Performance by Agent** (`stacked_bar_deal_performance.html`) — Shows how many deals each salesperson has at each stage, so you can compare workloads across the team.

Donut Charts (saved in the `donut` folder)

- **Opportunities by Project and Source** (`donut_project_source_indexedDB.html`) — A ring chart: the inner ring is which project a lead is interested in, the outer ring is where that lead came from.
- **Opportunities by Project and Salesperson** (`donut_project_user_indexedDB.html`) — Same idea, but the outer ring shows which salesperson is handling the lead instead.
- **Same two charts, with a "Download as Spreadsheet" button** (`csv_donut_project_source_indexedDB.html`, `csv_donut_project_user_indexedDB.html`) — Identical to the two above, with an extra button to export the numbers to a file you can open in Excel.

Funnel Charts (saved in the `funnel` folder)

- **Sales Funnel** (`funnel_widget_indexDB.html`) — Shows how many leads make it from one sales stage to the next — wider at the top, narrower at the bottom, like water through a funnel.
- **Current Stage Snapshot** (`funnel_current_stage.html`) — Instead of the flow-through view, just shows how many leads are sitting in each stage right now.
- **Simplified Funnel** (`funnel_one_chart.html`) — A condensed, single-chart version of the funnel above.

Reports (saved in the `report` folder)

- **Lead Aging Report** (`pipeline_aging.html`) — Shows how long leads have been sitting in each stage without being closed, grouped into buckets like "0-5 days" and "5-10 days," so you can spot leads going stale. It also checks whether a real person (not an automated message) actually replied to each lead.
- **Lead Aging Report — OIB North version** (`pipeline_aging_oib.html`) — Same report, customized for OIB North: only one salesperson can be picked at a time, there's no "Project" filter, and there's an extra "Source Category" filter.
- **Campaigns Report** (`campaign_report.html`) — Ranks marketing campaigns by how many leads each one brought in, with the full list of those leads underneath.
- **Pipeline Stages Report** (`enhanced_opportunities.html`, `opportunities.html`) — Two very similar reports showing how leads are spread across sales stages.
- **Opportunity Report (newer version)** (`opportunities_latest.html`) — A newer, improved redesign of the Pipeline Stages Report, with trend charts and a spreadsheet export button.
- **Lead Count by Agent** (`lead_count_by_agent.html`) — Counts how many leads each salesperson has, split between in-house marketing leads and outside-agency leads.

- **Lead Summary (Won Deals)** (`mkh_excel_report.html`) — Shows how many leads turned into "Won" deals per project, cross-checked against whether the salesperson actually messaged the lead back.
- **Funnel + Lead Audit Table** (`mkh_table.html`) — A funnel chart plus a searchable table of every lead underneath, so you can look up a specific one.
- **Opportunities by Project and Source (Report version)** (`project_source_report.html`) — A table version of the donut chart above, with a spreadsheet export button.
- **Opportunities by Salesperson and Stage** (`report_widget.html`) — A simple grid showing how many leads each salesperson has at each stage.
- **Sales Activity Report** (`sales_activity_report.html`) — Tracks whether salespeople are messaging and calling their leads every day, and flags leads nobody has contacted yet.
- **Sales Activity Report — two client versions** (`sales_activity_report_lsh.html` , `sales_activity_report_mkh.html`) — The same Sales Activity Report, each tweaked slightly for a specific client.

Call Charts (saved in the `call_chart` folder)

- **Call Status Report** (`call_chart1.html`) — Bar and pie charts showing how many calls were answered vs missed, for a chosen date range.
- **Client Call Data** (`call_report1.html`) — A plain table listing every call, with its status and who it was to/from.
- **Call Evaluation Scorecards** (`call_evaluation.php`) — A pass/fail scorecard for each call based on a checklist. Unlike the other call reports, this one still runs on the older server setup.

A couple of old, unused leftover files (`proxy.php` , `call_evaluation.py`) are also sitting in that folder from before we upgraded things — they're not used anymore and are safe to clean up later.

Where The Files Actually Live (The Technical Backend)

You don't need to understand this section to use any of the reports — it's here for whoever needs to fix something or add a new client later.

All the report pages are stored in an Amazon storage locker called **mnc-widgets-static**, and shown to the world through Amazon's fast-delivery service called **CloudFront**. Whenever someone changes a report page, that change has to be re-uploaded to this storage locker, or nobody will see the update.

The data (leads, pipelines, etc.) is fetched by small automatic programs called **Lambda functions** — think of them as tiny robot workers that wake up only when needed, do one job, and go back to sleep. We have four:

Robot worker	What it does
`ghl-config-api`	Looks up basic settings for a company, like its name.
`ghl-data-proxy`	The main one — fetches leads, pipelines, contacts, and conversations from GoHighLevel.
`call-data-proxy`	Fetches call records from our phone system provider (Avanser).
`ghl-cache-warmer`	Wakes up every 20 minutes on its own and refreshes the saved data for every company, so reports load fast.

Where fetched data is remembered for a while

To save time, fetched data is kept for a while in a filing system called **DynamoDB** — leads and pipelines for about 20 minutes, company settings for up to a day. After that time is up, the next request fetches fresh data automatically.

Where company passwords are kept

Every company's secret login key (so we can talk to their GoHighLevel account) is kept in a locked digital safe called **SSM Parameter Store** — never written directly into any file.

Where the report pages themselves are stored

All widget HTML files live in the Amazon storage locker **mnc-widgets-static**, delivered through CloudFront.

A few important habits already built in

- If fetching a huge company's data takes too long, the system never shows a half-finished, undercounted result as if it were complete — it quietly finishes the job in the background first.
- If GoHighLevel temporarily says "slow down," the system automatically waits and tries again on its own.

- On 23 Sep 2026, we added two small extra pieces of information (a contact's ID and email) to the data a lead carries — this makes it easier to correctly match a lead to the right person even if their phone number changed later.

Other Connected Systems (Not Part Of The Charts Above)

Beside the charts and reports, this project also quietly connects GHL to several other outside systems. Most of these run on one shared rented computer (a "Lightsail" server) — think of it as one filing cabinet that several smaller tools share space in.

- **crm-tracker** — A separate mini-dashboard showing GHL numbers for many companies at once.
- **WA** — A tiny tool that creates WhatsApp "click to chat" links. Nothing fancy, no logins involved.
- **bukku** — Automatically sends yesterday's invoices from our accounting software (Bukku) into GHL.
- **engage** — Connects GHL to a college admissions system (for Concord College). Hosted somewhere different from the others — we're not fully sure where yet.
- **IMU Integration** — Keeps contacts in sync between GHL and Zoho CRM (another company's customer database).
- **propertyx** — Forwards new leads into the Property-X system, and sends a Telegram alert if anything fails to go through.
- **twilio** — Handles incoming phone calls for about 17 different companies, routing each call based on which company it's for.
- **visitor_qr** — Sends WhatsApp check-in messages for visitors at Avaland's properties.
- **osk hana and osk shorea** — Two identical little relay programs that forward incoming messages along to GHL, with backup email alerts if something goes wrong.
- **scripts** — Leftover one-time setup tools used when we first moved everything to Amazon's cloud. Nobody needs to run these again.

A couple of things still on the old setup

The Call Evaluation scorecards (mentioned above) still run the old way, directly from the older shared server, instead of the newer Amazon cloud setup the rest of the call reports use.

Things Worth Knowing About (Clean-Up List)

- **Old passwords and keys need a check-up.** Several of the smaller connected systems above still have old login secrets stored directly in their files instead of in a safe digital vault. This should be cleaned up carefully by someone technical — not something to worry about day-to-day, just flagged so it doesn't get forgotten.
- **One connected system's exact home is unconfirmed.** We're not fully sure which server the college-admissions connector (`engage`) runs on — it looks different from all the others.
- **A couple of old unused files are still sitting around** in the call-charts folder from before the upgrade. They're not doing anything, just safe-to-delete leftovers.
- **The Call Evaluation scorecards haven't been upgraded yet** to the newer, faster cloud setup the other call reports use.
- **One report fetches its own data separately** instead of using the shared, already-fixed version — so future fixes to the shared version won't automatically apply to it.
- **The "Project" filter on the main Lead Aging Report may be empty for some companies** — the same kind of small bug already fixed for the "Source Category" filter on OIB North's version. Not fixed everywhere yet, since it may only affect some companies.